Silent Action Drift in OpenVLA: A Reproducible Prompt-Robustness Study on BridgeData V2

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Abstract

Vision-language-action (VLA) models promise a simple interface for robot control: the policy sees an image, reads an instruction, and predicts the next robot action. Yet recent work has shown that strong task success can coexist with weak language grounding, especially under contradictory or out-of-distribution instructions. This paper presents a fully reproducible empirical study of that problem in OpenVLA-7B using BridgeData V2 images and a local WSL inference pipeline. We evaluate two complementary branches: *Manual40*, a hand-curated set of 40 trajectories with task-specific prompts, and *Auto50*, an automatically sampled set of 50 trajectories from the scripted BridgeData split. For each image we compare four prompt families—normal, paraphrased, contradictory, and neutral—and measure the resulting action change using the first six action dimensions. Across both branches, paraphrases are consistently closest to the normal prompt, while contradictory and neutral prompts induce larger drift. Final-state images are more fragile than initial-state images in both branches. In the broader Auto50 study, mean 6D L_2 drift rises from 0.056 to 0.081 for paraphrases and from 0.070 to 0.094 for contradictions when moving from initial to final frames. The paper does not claim a complete safety solution. Instead, it contributes a rigorous real-image evaluation protocol, a practical local pipeline, and concrete evidence that action drift is a useful diagnostic signal for future failure anticipation and recovery mechanisms.

1 Introduction

Large vision-language-action (VLA) models offer an appealing story for robot learning: a single model can read an instruction, look at a camera image, and emit a low-level action for the robot to execute. OpenVLA is especially attractive in this space because it is open-source, inspectable, and designed to be fine-tuned or deployed outside a closed industrial stack [1, 5]. At the same time, recent literature has raised a serious concern. High benchmark success does not necessarily mean that a VLA is truly grounded in language. LangGap argues that strong VLA task performance can coexist with a substantial language gap [8]. IGAR shows that some models continue to produce visually plausible actions even when the instruction contradicts the scene, a failure mode termed *linguistic blindness* [9]. LangForce frames the same phenomenon as *information collapse*, where the instruction contributes too little additional information beyond the image and therefore stops shaping the output action reliably [12].

This paper studies that problem from a practical and reproducible angle. Instead of proposing a new VLA or a new training recipe, we ask a narrower question: *how much does the predicted OpenVLA action drift when the wording of the prompt changes while the image stays fixed?* We answer that question on real robot images rather than synthetic perturbations alone. The evaluation is built on BridgeData V2, a large public dataset of 53,896 trajectories collected across 24 environments [3]. We use both initial and final frames from trajectories because the model may behave differently at the start of a task than near task completion.

The resulting paper has three concrete contributions. First, it documents a working local evaluation

pipeline for OpenVLA-7B under WSL, including the attention-mask repair needed to make inference run reliably on commodity hardware. Second, it reports a two-branch real-image prompt-robustness study: a hand-curated **Manual40** branch (40 trajectories, 80 images total) and a broader **Auto50** branch (50 automatically sampled trajectories, 100 images total). Third, it shows a stable empirical pattern across both branches: paraphrases preserve behavior best, contradictory and neutral prompts induce larger drift, and final-state images are consistently more fragile than initial-state images. These results do not solve VLA safety. They do, however, provide a rigorous benchmark slice and a strong motivation for future inference-time detectors or recovery triggers.

2 Why this project matters

A household or factory robot cannot assume that every user will phrase an instruction in exactly the way seen during training. One person may say "put the cup in the sink," another may say "place the mug into the sink," and a third may give an incomplete or even contradictory command. If a robot policy changes its low-level action significantly under such wording shifts, the result can be a silent error: the robot still moves, but it moves for the wrong reason. That makes prompt robustness a safety and trustworthiness issue, not just a language issue [10, 11]. This project was designed to test that behavior directly, using real robot images and fixed prompts rather than anecdotal examples.

3 Related work

3.1 VLA models and the choice of OpenVLA

OpenVLA is a 7B-parameter open-source VLA trained on 970k real-world robot demonstrations [1]. Its openness, public checkpoints, and known fine-tuning recipes make it a natural model for reproducible experimentation [5]. We discuss π_0 in this paper because it is an important modern comparison point—a flow-matching VLA for general robot control built on a pre-trained vision-language model [2]. However, the empirical study in this paper evaluates OpenVLA only. We made that choice because the goal here is depth and reproducibility on a single model, not breadth across multiple architectures.

3.2 Prompt robustness, grounding, and failure detection

Recent work has increasingly argued that VLA evaluation must move beyond raw task success. LIBERO-X is a robustness benchmark designed to expose weaknesses hidden by easy language settings [7]. LangGap shows that current models can largely ignore language unless the benchmark forces genuine language understanding [8]. IGAR studies contradictory instructions and proposes train-free attention recalibration to restore instruction influence [9]. SAFE and FailSafe move the conversation toward safety-aware detection and recovery, but both rely on machinery beyond the frozen base policy itself [10, 11]. Q-DIG expands the failure surface by generating adversarial or diversity-seeking prompts for red-teaming [13]. Relative to that literature, our contribution is narrower but concrete: we provide a reproducible real-image study of prompt-induced action drift across initial/final states and across manual/automatic prompt construction.

3.3 Why BridgeData V2 and why the 30GB scripted split

BridgeData V2 provides large-scale real-robot trajectories with environment and task diversity [3]. For this project, using the official dataset mattered more than using random screenshots. The local scripted split occupied roughly 30 GB on disk, but that storage cost bought three benefits: trajectory-grounded prompts, reproducible frame selection, and a broader automatically sampled validation branch. Those benefits were important enough to justify the download.

4 Problem setup and metrics

Let x be a fixed image and let $p \in \mathcal{P}$ denote a prompt variant from the set

$$\mathcal{P} = \{p_{\text{norm}}, p_{\text{para}}, p_{\text{contra}}, p_{\text{neutral}}\}.$$

A frozen OpenVLA policy with parameters θ outputs a 7D action vector

$$a_\theta(x, p) \in \mathbb{R}^7.$$

Following the evaluation used throughout the project, we compare only the first six dimensions

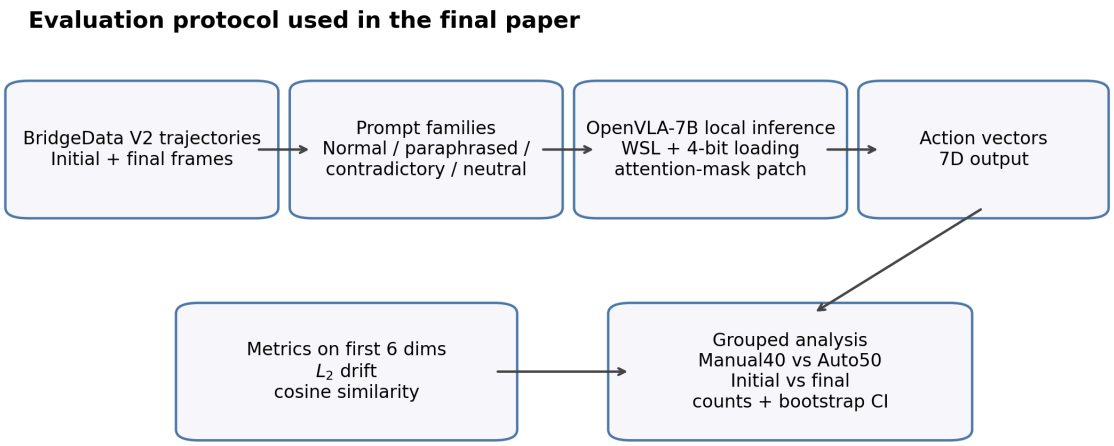

Figure 1: Evaluation protocol used in the final paper. Each image is paired with four prompt families. OpenVLA-7B is run locally with a repaired attention mask, and the first six action dimensions are compared using L_2 drift and cosine similarity. Results are aggregated separately by branch (Manual40 vs. Auto50) and by state (initial vs. final).

and exclude the final gripper channel because it was nearly constant in the collected runs. Let

$$\tilde{a}_\theta(x,p) = a_\theta(x,p)_{1:6}.$$

We then define prompt-induced drift relative to the normal prompt as

$$d_2(x,p) = \|\tilde{a}_\theta(x,p) - \tilde{a}_\theta(x,p_{\text{norm}})\|_2$$

and directional agreement as

$$s_{\cos}(x,p) = \frac{\tilde{a}_\theta(x,p) \cdot \tilde{a}_\theta(x,p_{\text{norm}})}{\|\tilde{a}_\theta(x,p)\|_2 \, \|\tilde{a}_\theta(x,p_{\text{norm}})\|_2}.$$

Smaller d_2 and larger $s_{\cos}$ indicate greater agreement with the normal prompt.

To compare fragility between early and late task states, we also use a simple state-gap summary:

$$\Delta_{\text{state}}(p) = \bar{d}_2^{\text{final}}(p) - \bar{d}_2^{\text{initial}}(p).$$

A positive value indicates that final frames are more sensitive than initial frames for that prompt family. For uncertainty reporting, we keep the original bootstrap confidence interval computation from the project bundle and report the resulting 95% intervals for key metrics.

5 Experimental protocol

5.1 Local inference pipeline

All experiments were executed locally in WSL using OpenVLA-7B in 4-bit mode on an RTX 2060 GPU. This was not a cosmetic choice. The project originally began as a lighter pilot in notebook environments, but stable evaluation required local control over dependencies, file layout, and checkpoint caching. The local pipeline therefore became part of the contribution: it demonstrates that a modern open VLA can be probed reproducibly without cloud-only infrastructure.

A key engineering issue emerged during inference. After visual tokens were fused into the model input, the cached text-only attention mask no longer matched the combined image-text sequence. The practical fix was to prepend a 256-token image mask before calling `predict_action`. In notation, if $M_{\text{text}} \in \{0,1\}^T$ denotes the original text mask, we replace it with

$$M' = [\mathbf{1}_{256}; M_{\text{text}}],$$

where $\mathbf{1}_{256}$ is an all-ones mask over the fused image-token prefix. This patch does not modify the learned weights; it only repairs inference-time alignment so the public checkpoint can be evaluated consistently on local hardware.

5.2 Three stages of evaluation

The full project unfolded in three stages.

Stage 1: 5-image initial-state pilot. The first pilot used five initial-state images only. Its role was debugging and pipeline validation, not scientific

evidence. We therefore report it in the appendix rather than mixing it into the main claims.

Stage 2: Manual40. We then built a hand-curated subset of 40 BridgeData trajectories. For each trajectory we saved one initial frame and one final frame, producing 80 images total. We wrote a task-specific normal instruction manually for each trajectory and derived paraphrased, contradictory, and neutral variants from it.

Stage 3: Auto50. Finally, we downloaded the scripted BridgeData split locally and created a broader automatic branch. We sampled 50 trajectories, extracted one initial and one final frame from each, and built prompt variants from the recorded run-family metadata. This produced 100 images. Taken together, the main study covers 180 image evaluations.

5.3 Prompt families

For each image we evaluated four prompt families:

1. **Normal**: the intended task instruction.
2. **Paraphrased**: meaning preserved, wording changed.
3. **Contradictory**: intentionally opposes the normal task.
4. **Neutral**: generic prompt without task-specific guidance.

The point of this design is to separate three different stress types: meaning-preserving lexical variation, meaning-breaking contradiction, and underspecification.

5.4 Why OpenVLA instead of π_0 in the main experiment

Table 1 explains the scope decision. π_0 is important in the literature and should be tested in future work, but the present study focused on a single open model to keep the pipeline auditable from end to end. The user requirement for a complete local run, the repaired inference path, and the trajectory-grounded BridgeData sampling would all have to be rebuilt for a second architecture. For this paper, empirical depth on one model was more valuable than a shallow two-model comparison.

6 Results

6.1 Manual40: prompt drift on a hand-curated subset

Table 2 reports the Manual40 results. The pattern is clean. In initial frames, paraphrased prompts are closest to the normal prompt under both metrics. Mean 6D L_2 drift is 0.028 for paraphrases, 0.042 for contradictions, and 0.045 for neutral prompts. In final frames the gap grows: 0.043 for paraphrases, 0.060 for contradictions, and 0.071 for neutral prompts. Cosine similarity tells the same story. Initial-frame cosine stays high for paraphrases (0.679) and falls for contradictory (0.586) and neutral (0.426) prompts. In final frames, the neutral mean drops slightly below zero, indicating that some directions of motion become almost unrelated to the normal prompt.

Counts reinforce the same conclusion. In Manual40, paraphrases are the closest variant in 25 of 40 initial frames and 22 of 40 final frames. Final frames also contain fewer near-invariant cases (4 versus 8), suggesting that late-stage scenes are less forgiving when the wording shifts.

6.2 Auto50: broader validation on the downloaded scripted split

The Auto50 branch preserves the same ordering under a broader and more heterogeneous sample. Initial frames show mean 6D L_2 drift of 0.056 for paraphrases, 0.070 for contradictions, and 0.073 for neutral prompts. Final frames again become more fragile, rising to 0.081, 0.094, and 0.082 respectively. Cosine similarities also drop from initial to final frames, especially under contradictory prompts. The wider branch therefore confirms that the Manual40 pattern is not a hand-curation artifact.

6.3 Cross-branch synthesis

Figures 2 and 3 make the global pattern easy to read. First, paraphrases are always the least disruptive prompt family. Second, final-state images are always more fragile than initial-state images under the same branch. Third, the automatically sampled branch is harder overall than the manual branch.

We can summarize the final-state fragility gap with $\Delta_{\text{state}}(p)$. In Manual40, Δ_{state} is 0.015 for paraphrases, 0.018 for contradictions, and 0.025 for

Table 1: Why OpenVLA was chosen for the main experiments, while π_0 is discussed as a follow-up comparison rather than as a co-evaluated model.

Model	Core formulation	Strength for this work	Why not the main paired baseline
OpenVLA	7B autoregressive open-source VLA [1]	Inspectable, public checkpoint, local WSL reproducibility, known bridge-style inference path	Chosen as the primary study model
π_0	Flow-matching VLA for general robot control [2]	Important comparison point in related work and future evaluation	Not run here; adding it would change the empirical scope and require rebuilding the local evaluation path

Table 2: Manual40 branch (40 trajectories, 80 images total). Means are computed on the first six action dimensions. Lower L_2 is better; higher cosine is better.

Group	L_2 drift			Cosine similarity		
	Para	Contra	Neutral	Para	Contra	Neutral
Initial	0.028	0.042	0.045	0.679	0.586	0.426
Final	0.043	0.060	0.071	0.480	0.219	-0.002

Table 3: Auto50 branch (50 trajectories, 100 images total). The pattern paraphrase < contradiction/neutral survives the move to a broader automatic subset.

Group	L_2 drift			Cosine similarity		
	Para	Contra	Neutral	Para	Contra	Neutral
Initial	0.056	0.070	0.073	0.484	0.269	0.171
Final	0.081	0.094	0.082	0.387	0.157	0.220

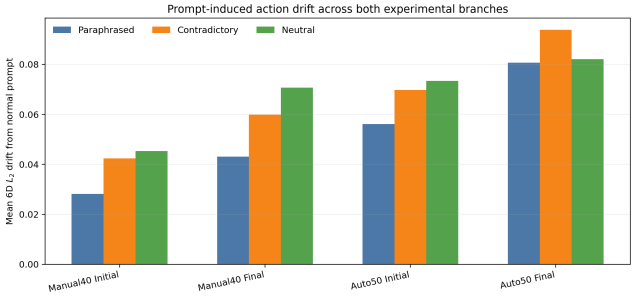

Figure 2: Mean 6D L_2 drift relative to the normal prompt. The same ranking appears in both branches: paraphrase is closest, while contradiction and neutral prompts move farther away. Final-state images are consistently more fragile.

neutral prompts. In Auto50, the corresponding gaps are 0.025, 0.024, and 0.009. The exact ranking of contradiction versus neutral changes in Auto50 final frames, but the key message remains: final states are systematically less stable.

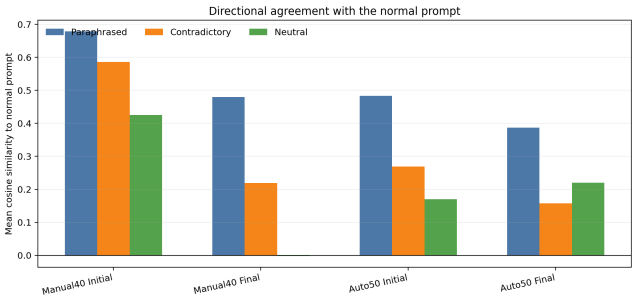

Figure 3: Mean cosine similarity to the normal prompt. Paraphrases preserve the strongest directional agreement, while contradictions and neutral prompts reduce alignment more aggressively.

Table 4: Counts-based robustness summary. "Para closest" means the paraphrased prompt had the smallest drift for that image among the three perturbed variants.

Branch	Group	Para closest	Contra farthest	Neutral farthest
Manual40	Initial	25/40	14/40	21/40
Manual40	Final	22/40	12/40	21/40
Auto50	Initial	27/50	15/50	22/50
Auto50	Final	26/50	16/50	15/50

6.4 Bootstrap confidence intervals

The project bundle also computed bootstrap confidence intervals. Figure 4 shows that the paraphrase-versus-contradiction/neutral ordering is not driven by a single outlier image. On the L_2 axis, the intervals stay separated enough to support the qualitative claim that paraphrases preserve the policy best, especially in the Manual40 branch. In the Auto50 branch, the contradiction-neutral gap is smaller, which matches the visual impression that the broader branch is noisier and less neatly ordered than the hand-curated one.

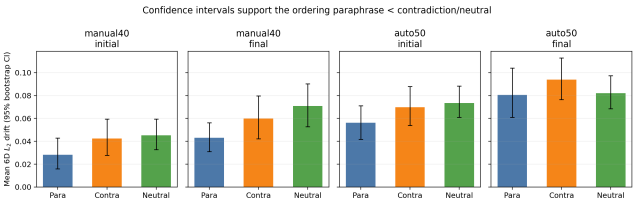

Figure 4: 95% bootstrap confidence intervals for the mean 6D L_2 drift. The paraphrase-vs.-contradiction/neutral ordering remains visible after uncertainty is added.

7 Discussion

7.1 What research gap does this paper actually solve?

This paper does not solve silent failure detection or language grounding in full. What it solves is narrower and still useful: it gives a reproducible, trajectory-grounded measurement protocol for *prompt-induced action drift* in a real open VLA under real robot images. The literature already says that VLA language grounding can be weak [8, 9, 12]. What was missing in the user's workflow was a stable local study that quantifies how much the action moves under four controlled prompt families, how that movement changes between initial and final states, and whether the result survives a move from manual curation to broader automatic sampling.

7.2 Why the results are useful

The practical value is straightforward. A VLA does not need to catastrophically fail on every contradictory instruction to be unsafe. It only needs to fail unpredictably. The present study shows that language sensitivity is not uniform; it depends on state and prompt family. That means future safety layers should probably be *state-aware* rather than global. A single threshold for all images may be too weak. A more realistic system would treat late-stage scenes as higher risk and monitor disagreement more closely there.

7.3 Why the local WSL story matters

The engineering story is not separate from the research story. In many student or lab environments, a result is only useful if another person can run it without access to specialized cloud infrastructure. This project required a full local chain: WSL, 4-bit model loading, patched attention masks, official dataset sampling, reproducible CSV output, and post-hoc analysis. That makes the work useful not just as a paper but also as a template for other low-resource labs.

7.4 Why OpenVLA and why not just compare everything at once?

The temptation in VLA research is to widen scope too quickly: compare multiple backbones, add simulation, add recovery, add red-teaming, and add failure detection all at once. This paper deliberately resists that temptation. OpenVLA was chosen because it is open, widely used, and compatible with local introspection [1]. π_0 remains an important comparison model, but running it well would require rebuilding parts of the inference stack. The right next step is therefore not a rushed second-model comparison. The right next step is a targeted intervention study built on the drift metrics already established here.

8 Limitations

This paper has six clear limitations. First, it evaluates *action drift*, not *task success*. We do not yet know how often a high drift score corresponds to a failed rollout. Second, the study uses one VLA family, OpenVLA, rather than a multi-model panel. Third, the contradictory and neutral prompts are designed, not crowdsourced; another prompt distribution might change magnitudes. Fourth, the manual and automatic branches are intentionally different in prompt construction, which makes them complementary but not identical. Fifth, the gripper dimension was excluded from the main metrics because it was nearly constant in the collected runs. Sixth, the local attention-mask repair is an implementation patch, not a theoretical model contribution.

9 Future work

The most valuable next experiment is an **outcome-aware safety layer**. The present paper already provides the drift measurements needed to motivate it. A natural follow-up is to define a detector $\delta(x, p)$ that triggers when cross-prompt disagreement exceeds a threshold, then evaluate whether the trigger predicts actual task failure in simulation. This is

where SAFE and FailSafe become especially relevant baselines [10, 11]. Another strong extension is a **cross-model comparison** between OpenVLA and π_0 under the same BridgeData frame protocol. A third extension is a **real or simulated recovery policy**, such as re-prompting, abstaining, or performing a safe retreat. Finally, it would be valuable to connect this frame-level work to newer robustness benchmarks such as LIBERO-X and contradictory-instruction settings such as ICBench [7, 9].

10 Conclusion

This paper presented a fully reproducible study of prompt-induced action drift in OpenVLA-7B using BridgeData V2 images and a local WSL evaluation pipeline. The study combined a 5-image debugging pilot, a 40-trajectory manual branch, and a 50-trajectory automatic branch. Across both main branches, the same pattern held: paraphrases preserved behavior best, contradictory and neutral prompts induced larger drift, and final-state images were consistently more fragile than initial-state images. The result is not a final safety mechanism, but it is a strong empirical base for one. More broadly, it shows that prompt robustness in VLA systems can be studied rigorously with open models, official robot datasets, and modest local hardware—which is exactly the kind of research infrastructure the field needs.

Acknowledgments

This project was carried out by Harmit Patel (student ID 110164115), Department of Computer Science, University of Windsor, Windsor, Ontario, Canada. The work was completed under the supervision of Sheriff Saad. **For actual CoRL submission, this named page should be replaced by the anonymous version included in the package.**

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A Appendix A: Pilot study naming

The smallest pre-main-study experiment in this project used five initial-state images only. In the write-up, the correct name for that experiment is *5-image initial-state pilot experiment*. It should not be described as a full-trajectory study.

B Appendix B: Key exact numbers

Manual40 initial means: paraphrased 0.028126, contradictory 0.042365, neutral 0.045330. Manual40 final means: paraphrased 0.043141, contradictory 0.059920, neutral 0.070772. Auto50 initial means: paraphrased 0.056164, contradictory 0.069716, neutral 0.073405. Auto50 final means: paraphrased 0.080668, contradictory 0.093857, neutral 0.082103.

C Appendix C: Suggested supplementary video

No true rollout videos were included in the uploaded artifact set, so this package does not fabricate them. The recommended supplementary format is to show one baseline rollout and one monitored rollout per high-drift case, with the prompt text shown on screen. A short README for that supplementary structure is included in the package.

D Appendix D: Example prompt construction

Table 5 shows representative prompts from the study. The goal was not to generate adversarial language with a separate LLM. The goal was to keep the perturbations human-readable and operationally plausible. A household user, a factory operator, or a data annotator could plausibly produce all four prompt families used here.

E Appendix E: Selected confidence intervals

Table 6 reproduces the bootstrap intervals used in the paper package. The main body only needed the qualitative takeaway. This appendix keeps the exact numbers available for readers who want stronger statistical detail.

F Appendix F: Why OpenVLA first and π_0 next

The project repeatedly returned to the same practical choice: whether to broaden the study across models or finish one rigorous evaluation on a single inspectable model. OpenVLA won that choice because the local WSL path was already working, BridgeData-style prompting was already wired to its processor, and the attention-mask repair had already been validated. This appendix figure does not claim that OpenVLA is the best model overall. It explains why it was the correct model for this particular paper.

Model	Core formulation	Strength for this work	Why not main experiment
OpenVLA	7B autoregressive open-source	Inspectable, local reproducibility, existing	Chosen as main study model
π0	Flow-matching VLA for general	Important comparison point in related wo	Not run in this paper; adding it now would c

Figure 5: Conceptual comparison used to scope the paper. OpenVLA was selected because it was the model already running locally with a repaired inference path. π_0 remains the most natural second-model extension.

G Appendix G: What would make this a stronger CoRL paper

A stronger CoRL submission would add one of the following extensions without changing the existing core:

1. **Outcome-aware validation**: replay high-drift cases in simulation and measure task success rather than action disagreement alone.
2. **Runtime drift detector**: treat cross-prompt disagreement as an inference-time warning signal and trigger re-prompting, abstention, or safe retreat.
3. **Cross-model evaluation**: rerun the same protocol on π_0 or another open VLA to separate model-specific from model-general behavior.
4. **Structured contradictory benchmark**: reuse the prompt families from this paper but execute them in a benchmark closer to ICBench or LIBERO-X.

These are promising because they build directly on the current data and code instead of discarding the existing work. That is also why the present paper is still worth keeping: it establishes a clean

Table 5: Representative prompt construction used in the study. The exact wording varied by trajectory, but the structure was consistent.

Trajectory task	Normal prompt	Paraphrased prompt	Contradictory prompt	Neutral prompt
Put the pot on the towel	In: What action should the robot take to put the pot on the towel? Out:	In: What should the robot do to place the pot on the towel? Out:	In: What action should the robot take to avoid the pot and move away from the towel instead? Out:	In: What action should the robot take? Out:
Fold the cloth from top right to bottom left	In: What action should the robot take to fold the cloth from top right to bottom left? Out:	In: What should the robot do to fold the cloth from top right to bottom left? Out:	In: What action should the robot take to avoid folding the cloth and move away instead? Out:	In: What action should the robot take? Out:
Pick up the rigid object	In: What action should the robot take to pick up the rigid object? Out:	In: What should the robot do to pick up the rigid object? Out:	In: What action should the robot take to avoid the target object and move away instead of trying to pick it up? Out:	In: What action should the robot take? Out:

baseline that later interventions can be measured against.

H Appendix H: Reproducibility checklist

For completeness, the concrete artifact trail for this paper is listed here.

1. **Model**: OpenVLA-7B loaded from the public Hugging Face/OpenVLA release in 4-bit mode.
2. **Runtime**: local WSL environment with CUDA-enabled PyTorch on an RTX 2060 GPU.
3. **Patch**: prepend a 256-token image mask before `predict_action`.
4. **Manual branch**: 40 trajectories, one initial and one final frame per trajectory, manually authored prompt families.
5. **Automatic branch**: 50 scripted BridgeData trajectories, one initial and one final frame per trajectory, prompt families derived from run-family metadata.
6. **Metrics**: first six action dimensions only; mean 6D L_2, cosine similarity, closest/farthest counts, and bootstrap confidence intervals.
7. **Outputs**: CSV result files, metadata files, summary plots, and both named/anonymous LaTeX sources.

This checklist is included because one of the practical goals of the project was not only to report a result but also to leave behind a complete package that another student, reviewer, or labmate could re-open later without reconstructing the debugging process from memory.

Table 6: 95% bootstrap confidence intervals for the mean 6D L_2 drift.

Experiment	Group	Metric	Mean	95% CI
Manual40	Initial	L_2(normal, paraphrased)	0.028126	[0.016002, 0.042951]
Manual40	Initial	L_2(normal, contradictory)	0.042365	[0.027787, 0.059539]
Manual40	Initial	L_2(normal, neutral)	0.045330	[0.032649, 0.059542]
Manual40	Final	L_2(normal, paraphrased)	0.043141	[0.030908, 0.056137]
Manual40	Final	L_2(normal, contradictory)	0.059920	[0.042288, 0.079732]
Manual40	Final	L_2(normal, neutral)	0.070772	[0.052691, 0.090367]
Auto50	Initial	L_2(normal, paraphrased)	0.056164	[0.041880, 0.071180]
Auto50	Initial	L_2(normal, contradictory)	0.069716	[0.053953, 0.087871]
Auto50	Initial	L_2(normal, neutral)	0.073405	[0.060928, 0.088257]
Auto50	Final	L_2(normal, paraphrased)	0.080668	[0.061054, 0.103965]
Auto50	Final	L_2(normal, contradictory)	0.093857	[0.076429, 0.112974]
Auto50	Final	L_2(normal, neutral)	0.082103	[0.064974, 0.101565]